

Multidirectional Superimposed Projection for Delay-free Shadow Suppression on 3D Objects

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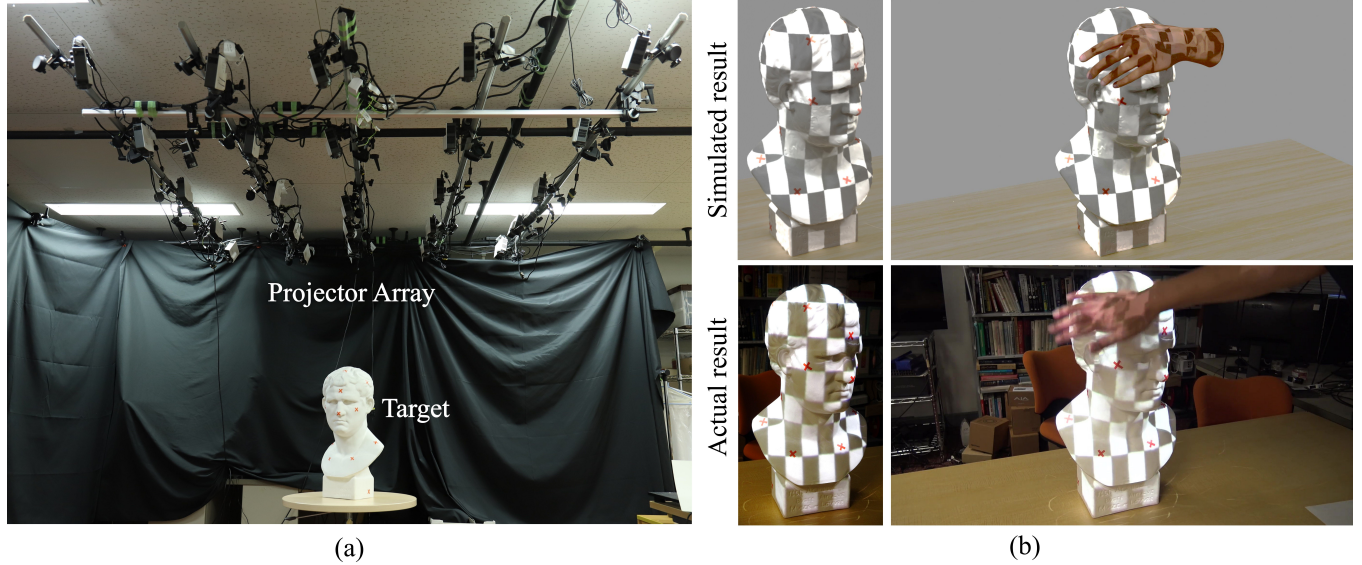


Figure 1: (a) Setup of our projection system. (b) Simulation results (top row) compared with projection results using our multidirectional superimposed projection method (bottom row). Left: Results of projecting a checker pattern onto the target object. Right: Results when holding the hand over the object during projection. The shadow of the hand does not clearly appear.

Abstract

The method of superimposed projection, which involves using a projector array, allows for projection from various directions. This technique helps to minimize the appearance of shadows caused by obstacles. This paper explores the implementation of a multidirectional superimposed projection system and analyzes the projection results through both actual projection and simulation.

CCS Concepts

• **Computing methodologies** → **Mixed / augmented reality.**

Keywords

Projection mapping, mixed reality

ACM Reference Format:

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1 Introduction

Projection Mapping (PM) uses projectors to superimpose virtual images onto real-world objects, allowing for the editing of the object's appearance by projecting computer-generated texture images. Since users do not need to wear devices like head-mounted displays, PM is widely used in entertainment, education, medical settings, and more [Nishino et al. 2018]. However, one issue with PM is that the projected light can be blocked by the user's body or other objects, creating shadows in the projected image. This limitation restricts user behavior, such as preventing them from crossing in front of the projector. Additionally, if the projection surface causes specular reflection, users may perceive a glare in the appearance provided by the PM. These issues can result in a discrepancy between the projected result and the intended appearance.

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As a solution to these problems, a method using cameras to detect shadows and increasing the illumination in shadowed areas with multiple projectors to eliminate shadows has been developed [Summet et al. 2007]. However, this method inevitably incurs latency when projecting onto dynamic occluders as it involves shadow removal processing after detecting shadows. Another method estimates the shadow region by tracking the occluders and dynamically removing the shadows [Audet and Cooperstock 2007]. Nonetheless, this approach also introduces latency during the estimating process, making it challenging to accurately remove shadows from fast-moving occluders.

These methods inherently involve computational processing, which can cause delays. Therefore, we propose an optical approach that eliminates the need for computational processing, thereby avoiding any delays. One such optical approach uses a retrotransmissive plate to project images from the projector onto the target surface with a wide viewing angle [Hiratani et al. 2023]. However, this method requires an object that is symmetric to the projection target, imposing restrictions on the size of the projection target. To address these limitations, we propose a projection system, as shown in Figure 1(a), comprising 25 projectors and cameras. This system can project onto objects of any shape.

2 Method

In our system, the same object is projected from multiple directions using 25 projectors. This multidirectional projection approach ensures that shadows do not occur as long as not all projection lights are blocked. Since this method does not require image processing for shadow removal, there is no latency introduced. Additionally, compared to conventional single-projection methods, this system reduces the brightness contribution on the projection surface from each projector. As a result, it minimizes the occurrence of specular reflection relative to the user's viewpoint.

To enable superimposed projection, our system is designed to position the projection target within the projection range of each projector. The projectors are arranged around the target, and multiple cameras are also positioned to encircle the target. We calibrate the system by projecting structured light patterns, known as gray code patterns, from each projector and capturing them with the cameras in the system. The decoding results of the captured images are used to calculate intrinsic and extrinsic parameters. Using these parameters, the projected images from each projector are rendered so that the projected result from each projector aligns with the target surface.

3 Result

The results obtained using the proposed method are shown in the bottom row of Figure 1(b). The projection target used was a bust ($0.32 \times 0.574 \times 0.261$ m) depicted in Figure 1(a). Red cross markers were attached to the surface of the projection target to establish correspondence between the 3D world coordinates and 2D projector coordinates of each projector. It can be observed that shadow formation on the surface of the target is suppressed even during occluding actions, such as placing a hand over the projection. Additionally, the simulation results of the projection using this system are displayed in the top row of Figure 1(b). The results

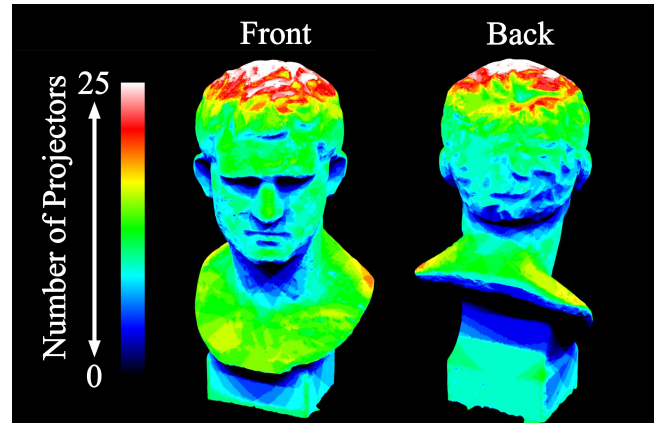


Figure 2: Contribution of the multidirectional projection, as visualized by the simulator.

show a high degree of similarity to the actual projection. Using this simulator, we generated a heatmap that displays the contribution of each projector at all points on the surface of the object during projection (Figure 2). The heatmap indicates that shadows are more likely to form on areas such as the cheeks and under the chin, while shadow formation is suppressed on the top of the head and forehead. This heatmap can be used in the future to optimize the arrangement of the projectors.

4 Conclusion

In this paper, we proposed a multidirectional superimposed projection system designed to suppress shadow formation without any delay, ensuring a seamless experience for users. We confirmed, through actual projection tests, that shadows were effectively suppressed even when the shielding object is positioned above. Simulations further demonstrated the success of superimposed projection.

Acknowledgments

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